Operational Dashboard for Battalion Chiefs

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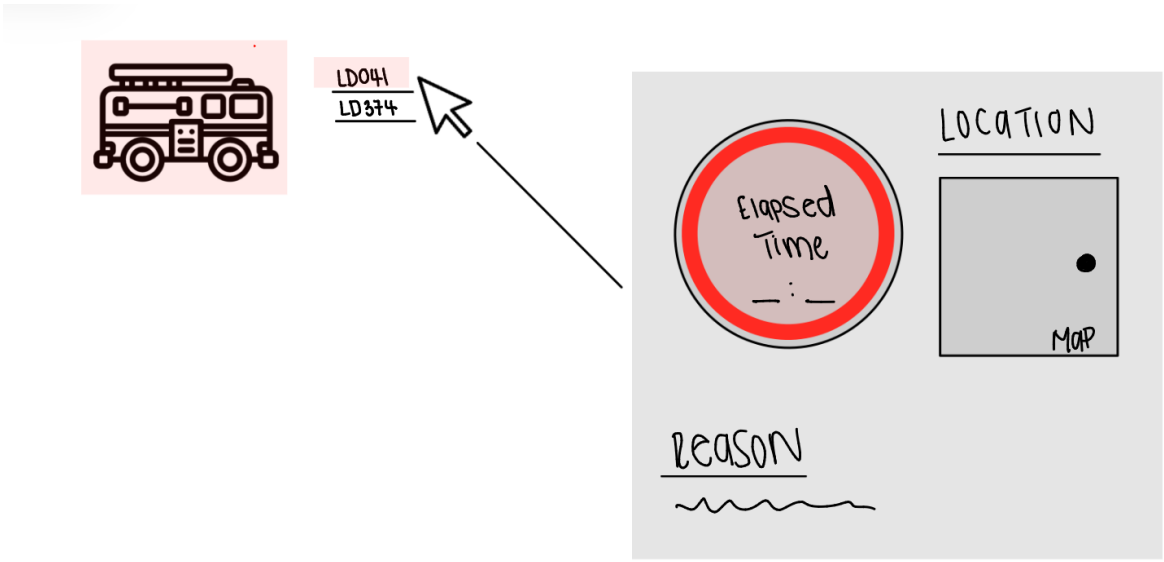

Fig. 1. A sketch of one of my ideas for the proposal

Abstract— Fire and Emergency Medical Services leadership rely on timely and accurate information to make operational decisions that affect public safety and responder workload. This project proposes an interactive, shift aware dashboard for Battalion Chiefs that presents real time operational metrics across fire and EMS units. The dashboard consolidates multiple key performance indicators, including turnout time compliance, overdue incident reports, out of service units, ambulance utilization, and time critical STEMI responses, into a single operational view.

A central design challenge addressed by this project is the definition of a Battalion Chief work shift. Shifts begin and end at 08:00 and span two calendar days, which complicates standard date based aggregation. The proposed visualization models shifts explicitly, ensuring that displayed data aligns with the active shift window. The design emphasizes temporal accuracy, comparative performance, and anomaly visibility in order to support situational awareness and short term operational decision making during an active shift.

Index Terms—Operational dashboards, Emergency medical services, Fire services, Real-time monitoring, Performance metrics

1 Introduction

Fire and Emergency Medical Services agencies generate large volumes of operational data during each shift, including dispatch times, unit availability, workload indicators, and patient care outcomes. Battalion Chiefs supervise multiple fire and ambulance units across a defined geographic area and remain responsible for monitoring these conditions while making operational decisions in real time. These decisions affect response performance, personnel workload, and overall system readiness. In practice, much of the relevant data is distributed across separate systems, static reports, or tabular outputs that do not support rapid situational assessment [5].

This project proposes the design of an interactive dashboard intended for use by Battalion Chiefs during an active shift. The dashboard integrates multiple operational metrics, including turnout time compliance, overdue incident reports, out of service units, ambulance unit hour utilization, and STEMI response performance, into a single interface. The design prioritizes live monitoring and short term decision support rather than retrospective analysis [6].

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The definition of a work shift presents a significant design constraint. Battalion Chief shifts run from 08:00 to 08:00 and span two calendar days. Conventional date based filtering introduces errors by including data outside the active shift while excluding relevant events within it. This visualization treats the shift as a first class temporal entity so that all displayed metrics reflect the correct operational window [6].

The primary users of this system are the Battalion Chiefs. Secondary users may include command staff or training officers reviewing operational performance. The need for this visualization arises from the complexity of managing multiple units simultaneously while balancing workload, maintaining compliance with performance standards, and responding to high risk incidents. The design applies established information visualization principles, including visual hierarchy, threshold based encoding, and coordinated views, to support accurate interpretation and timely decision making in a high consequence operational environment.

2 Approach

The visualization approach focuses on supporting rapid comprehension and operational awareness during an active Battalion Chief shift. The dashboard is organized around common analytical tasks per-

formed by Battalion Chiefs, including monitoring response time compliance, identifying unit availability issues, and tracking time sensitive medical incidents.

Key questions the visualization is designed to support include:

- Are units meeting turnout time performance targets for fire and EMS calls during the active shift?
- Are there overdue incident reports that require immediate follow up?
- Which units are currently out of service and for how long?
- Are ambulance units becoming overutilized or underutilized?
- Are STEMI responses meeting required time benchmarks?

Interaction is structured around a shift based temporal filter that defines a fixed 24 hour window from 08:00 to 08:00. This approach avoids reliance on calendar dates. Coordinated views allow filtering by battalion and unit when appropriate, while system wide views such as out of service units remain unfiltered to preserve awareness across battalions.

3 Initial Ideas

The following design concepts explore different ways of representing operational data for Battalion Chiefs. Each idea emphasizes a different aspect of the problem space, including real time system status, performance summary, and accountability. Together, these concepts examine tradeoffs between overview and detail, aggregation and specificity, and immediate awareness versus follow up action.

3.1 Idea 1

This design visualizes out of service units using iconography and color to support rapid system awareness. Each unit is represented by a stylized outline icon corresponding to its unit type, such as engine, ambulance, or ladder. This allows unit roles to be identified at a glance.

Icons appear alongside unit identifiers, with color encoding the duration of the out of service status. As elapsed time crosses predefined thresholds, the icon and corresponding identifier change color and visual emphasis. This draws attention to units that may require intervention.

Additional detail is available through interaction. Hovering over a unit identifier reveals contextual information including elapsed out of service time, location, and reason. This approach preserves a clean overview while supporting focused inspection when needed.

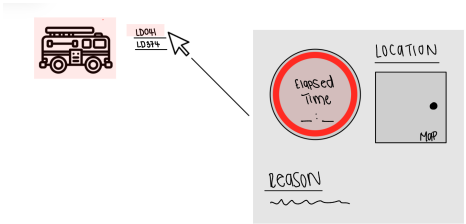

Fig. 2. Sketch 1: OOS View - Ladder Example

3.1.1 Pros

This design supports rapid situational awareness through the use of icons and color as preattentive signals. Unit types remain visually distinct, and time based color changes highlight prolonged outages without requiring textual scanning. The hover interaction provides a clear path from overview to detail. The system wide scope supports cross battalion awareness, which reflects actual operational practice.

3.1.2 Cons

The design relies heavily on color encoding, which requires careful palette selection to maintain accessibility. The view emphasizes current status and does not inherently communicate historical patterns. High numbers of out of service units may increase visual density without additional aggregation or filtering.

3.2 Idea 2

This design summarizes turnout time compliance using a unit based stacked bar chart. Each horizontal bar represents the total number of tracked turnout events for a unit within a selected time window. Bars are divided into MET and MISSED segments, enabling comparison across units while also conveying call volume.

An optional summary view presents the overall percentage of compliant turnouts for the selected period. The time window is adjustable and supports views such as the last 24 hours or multiple prior shifts.

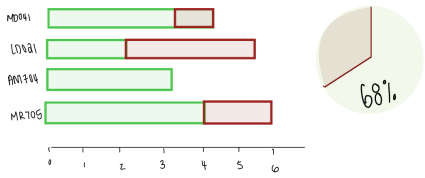

Fig. 3. Sketch 2

3.2.1 Pros

This approach provides a familiar and readable summary of turnout performance by unit. The stacked bars communicate both compliance and volume, which helps distinguish isolated misses from persistent issues. Time window controls support both short term monitoring and broader review. Sorting by compliance metrics supports prioritization.

3.2.2 Cons

Aggregation obscures temporal patterns within the shift. All missed turnouts receive equal visual weight regardless of severity. The design is conventional and may offer less expressive capacity than icon based or temporal views.

3.3 Idea 3

This design visualizes overdue incident reports using a station centered icon layout. Stations appear vertically on the left, with report icons displayed to the right. Each icon represents a single overdue report and uses color to indicate the responsible shift.

Interaction provides access to detail. Selecting a report reveals information about the responsible crew member, station, and elapsed time since the unit returned to service. This supports direct follow up while keeping the primary view compact.

An optional secondary view summarizes overdue reports by individual using a circular ring display. Interaction between views allows corresponding reports to be emphasized, supporting both station level and individual level assessment.

3.3.1 Pros

The design supports accountability by linking overdue reports to stations, shifts, and individuals. The document icon provides an intuitive visual metaphor. Interaction supports detail on demand without cluttering the overview. The optional individual focused view supports workload assessment at the personnel level.

3.3.2 Cons

High report counts may increase visual density and require aggregation or scrolling. Shift encoding through color requires careful design for accessibility. Temporal trends are not directly visible without interaction.

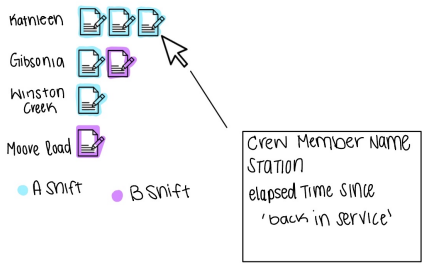

Fig. 4. Sketch 3

4 Progress Update

This section summarizes progress since the proposal stage, including refinement of the initial dashboard sketches, implementation of a working visualization prototype, integration of supporting research into the design approach, and current development challenges.

4.1 Refinement of Initial Sketch

During the proposal stage, several early layout concepts were explored for representing Battalion Chief operational conditions. These included icon based views for out of service unit tracking, stacked bar summaries for turnout compliance, and station centered layouts for overdue incident reports. Each of these sketches was designed around the types of monitoring tasks performed during an active Battalion Chief shift rather than general performance reporting.

Based on these initial concepts, the dashboard design was refined into a coordinated overview layout that integrates multiple operational indicators into a single interface. The refined structure focuses on turnout compliance, out of service unit tracking, overdue incident reports, ambulance unit hour utilization, and STEMI response timing. These components reflect the primary monitoring responsibilities identified earlier in the approach section.

An important design decision that remained consistent from the original sketches is the use of a shift based temporal window. Battalion Chief shifts run from 08:00 to 08:00 and span two calendar days, so the visualization treats the shift as a primary time structure rather than relying on calendar based filtering.

4.2 Working Visualization Prototype

A working prototype of the dashboard has now been implemented using HTML, CSS, JavaScript, and Chart.js. The prototype loads structured operational datasets and displays them through coordinated visualization panels aligned with the original design goals.

The current prototype includes:

- KPI summary cards showing turnout compliance, unit availability status, utilization conditions, and overdue reporting activity
- a stacked turnout compliance bar chart showing met and missed turnout events by unit
- a bar chart representing ambulance unit hour utilization
- tables displaying out of service units
- tables displaying STEMI response timing cases
- tables displaying overdue incident reports
- filtering controls for shift and battalion level views

The dashboard successfully loads processed sample data and presents multiple operational indicators within a single overview oriented interface intended to support situational awareness during an active shift.

4.3 Integration of Research Into the Design Approach

The dashboard design continues to follow principles identified in operational dashboard and emergency visualization research. Prior work shows that dashboards are most effective when they support real monitoring tasks, present clearly defined indicators, and avoid unnecessary visual complexity that increases cognitive load [1].

Research on emergency department throughput dashboards and emergency operations center visualization tools supports the use of overview oriented layouts that allow users to monitor system conditions without interrupting workflow [3, 9]. These findings influenced the decision to organize the interface around summary indicators rather than detailed analytical interaction.

Visualization research in emergency management also emphasizes the importance of supporting a shared operational picture across multiple system conditions [7, 2]. This directly informed the decision to combine turnout compliance, availability status, overdue reporting activity, utilization measures, and time critical medical response indicators into a single coordinated dashboard interface.

Studies of interactive analytical dashboards also suggest that excessive interaction can reduce situation awareness in operational environments, supporting the use of restrained interaction design in command level systems such as this one [4]. Additional lessons from public health dashboard development highlight the importance of clear metric definitions and consistent data practices when integrating multiple operational indicators into a single display [8].

4.4 Initial Results and Current Challenges

Initial prototype development shows that multiple operational datasets can be integrated into a coordinated dashboard layout that supports several key Battalion Chief monitoring tasks. Using sample Polk County Fire Rescue data, the visualization currently supports comparison of turnout compliance across units, identification of out of service resources, monitoring of overdue reporting activity, and review of ambulance utilization and STEMI timing performance.

Several implementation challenges remain.

Some datasets use different field naming conventions for battalion and shift identifiers, which makes coordinated filtering across panels more difficult than expected. In addition, some datasets are currently aggregated before visualization, which limits the effectiveness of shift based filtering behavior.

Because Battalion Chief shifts span two calendar days, additional preprocessing logic is also required to ensure that filtering reflects the correct operational window rather than calendar based date ranges.

Next development steps include improving shift aware filtering across datasets, standardizing battalion level identifiers between sources, refining coordinated panel interaction behavior, and preparing representative monitoring tasks for evaluation of the visualization.

5 Results

The approach will be validated through task based evaluation and qualitative feedback. A set of representative tasks will be defined, including identifying prolonged out of service units, locating overdue reports, and assessing turnout compliance. Participants will complete these tasks using the proposed visualization and a baseline method such as static email reports.

Task completion time and accuracy will be measured to assess efficiency and error reduction. Faster and more accurate performance will indicate improved operational support. Qualitative feedback will be collected through short surveys focused on clarity, usability, and perceived effectiveness in understanding battalion status.

6 Conclusion

The goal of this project is to develop a clear and interactive dashboard that supports Battalion Chiefs and EMS leadership in maintaining operational awareness across multiple domains. The design focuses on real operational tasks and reflects the constraints of shift based emergency services work. The proposed ideas explore different approaches to balancing overview and detail while maintaining clarity in a time sensitive environment.

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